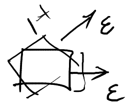
$\propto \sigma_{ij}$

$\propto \varepsilon_{ijk}$

$v = Au$

$\underline{\sigma} = \underline{\underline{C}} : \underline{\varepsilon}$ → no depende de la deformación

→ 36 componentes → 2 componentes




$\sigma_{ij} = C_{ijkl}\,\varepsilon_{kl}$

$E, \nu$

$\lambda, G \text{ o } \mu$

$K$

$$\underbrace{\sigma_{ij} = \lambda e_{\alpha\alpha}\delta_{ij}}+2\mu e_{ij}$$

$$\sigma_{ij} = C_{ijkl}\,\varepsilon_{kl} = \left[\lambda\delta_{ij}\delta_{kl} + \mu(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})\right]\varepsilon_{kl}$$

$$= \lambda\delta_{ij}\varepsilon_{kk} + \mu(\varepsilon_{ij} + \overset{\varepsilon_{ij}}{\varepsilon_{ji}}) = \lambda\delta_{ij}\varepsilon_{kk} + 2\mu\varepsilon_{ij}$$

$$\sigma_{ii} = \lambda e_{\alpha\alpha}\overset{3}{\delta_{ii}} + 2\mu \overset{e_{\alpha\alpha}}{e_{ii}} = (3\lambda + 2\mu)e_{\alpha\alpha} \Rightarrow$$

$\sigma_m = 1/3\,\sigma_{ii}$

$$\underbrace{e_{\alpha\alpha}} = \frac{\sigma_{ii}}{(3\lambda+2\mu)} = \frac{3\sigma_m}{3\lambda+2\mu}$$

→ deformación volumétrica por unidad de volumen

$$\sigma_{ij} = \lambda\frac{\sigma_{ii}}{(3\lambda+2\mu)}\delta_{ij} + 2\mu e_{ij} \Rightarrow e_{ij} = \frac{1}{2\mu}\left[\sigma_{ij} - \frac{\lambda\sigma_{ii}\delta_{ij}}{(3\lambda+2\mu)}\right]$$

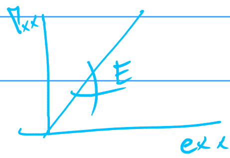
$$\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)} \qquad G = \mu = \frac{E}{2(1+\nu)}$$

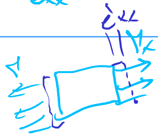
$$* \quad e_{ij} = \frac{1}{E}\left[\sigma_{ij} - \nu\left(\sigma_{\alpha\alpha}\delta_{ij} - \sigma_{ij}\right)\right]$$

$$e_{xx} = \frac{1}{E}\left[\sigma_{xx} - \nu\underbrace{(\sigma_{\alpha\alpha} - \sigma_{xx})}_{\sigma_{xx}+\sigma_{yy}+\sigma_{zz}-\cancel{\sigma_{xx}}}\right] = \frac{1}{E}\left[\sigma_{xx} - \nu(\sigma_{yy} + \sigma_{zz})\right]$$

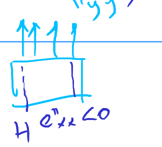
$$e_{xy} = \frac{1}{E}\left[\sigma_{xy} + \nu\sigma_{xy}\right] = \frac{1+\nu}{E}\sigma_{xy}$$

$$e_{xx} = \underbrace{\frac{\sigma_{xx}}{E}}_{e'_{xx}} - \underbrace{\frac{\nu}{E}\sigma_{yy}}_{e''_{xx}} - \frac{\nu}{E}\sigma_{zz}$$




$+$




$+$ caso z

De *

$$i \neq j \Rightarrow \delta_{ij} = 0 \Rightarrow e_{ij} = \frac{1+\nu}{E}\sigma_{ij} \quad \begin{cases} \cdots \\ \cdots \\ \cdots \end{cases}$$

$$i = j \Rightarrow \delta_{ij} = 1 \Rightarrow e_{ij} = \frac{1}{E}\left[\sigma_{ij} - \nu\left(\sigma_{\alpha\alpha} - \sigma_{ij}\right)\right] \quad \begin{cases} \cdots \\ \cdots \\ \cdots \end{cases}$$